

Employee Burnout Cluster 2026

Download the employee_burnout_turnover_prediction.zip from Canvas. Use mongoimport command to import the data into the employee_burnout collection of the employeesdb database with the following syntax:

```
C:\Users\jing>C:\Users\jing\Downloads\mongodb-database-tools-windows-x86_64-100.14.0\mongodb-database-tools-windows-x86_64-100.14.0\bin\mongoimport.exe --db=employeesdb --collection=employee_burnout --file=C:\Users\jing\Dropbox\cit\nosql\dataset\synthetic-employee-dataset.json --jsonArray
```

Complete the following tasks and screenshot each command and its output.

Stress level: high -> 76% – 100%

medium -> 51% – 75%

low -> 26% – 50%

resting -> 0% – 25%

Burnout risk level: high -> 76% - 100%

moderate -> 50% - 75%

low -> 0% - 49%

Part 1 (Self-Deployed MongoDB Cluster)

Our MongoDB cluster consists of 12 nodes (or independent servers), distributed over four small data centers across four different regions of USA (north, south, east, west). Each data centre represents a shard of the cluster, in which all nodes are organized as a replica set with the minimum configuration of a master-slave architecture. The configuration server has a group of three nodes {cfg0, cfg1, cfg2} organized as a replica set. The cluster is interfaced to the potential clients via lightweight processes. On one hand, these processes abstract and hide from clients the complexity of the cluster storing the data. The processes present the entire cluster as a single logical node, abstracting the actual physical distribution of the data among the shards. On the other hand, the processes also provide access to the metadata. The cluster is “simulated”, in the sense that it is built up on top of a single physical machine.

A script-based procedure is provided to allow you to build up the cluster on your own physical machine. To set up a distributed environment to store the diabetes application data in this self-deployed MongoDB cluster. Write bat (or bash) files and javascripts to set up this distributed cluster on a local machine.

Once the cluster is fully set up, we are ready to inject the diabetes application data and to distribute the data. A MongoDB database can be created on the cluster named burnoutAnalysisDB. The database can be sharded using a shard key. You need to be able to execute the scripts to set up the cluster and to demonstrate the whole process

of distributing data. Furthermore, you also need to explain what is happening when the scripts are executing.

To answer the following queries with the built cluster:

1. List the average salary and average duration of working contract for the laid off employees in each department where those employees had over 70% chances of making a career progression and had over 70% satisfaction rate for the department.
2. Which department has the largest number of employees laid off?
3. List out all teams in every department that have high sentiment score (i.e. above 60%) in which members attend meetings with high frequency (i.e. over 70%) with resting stress level and low burnout risk

Part 2 (Atlas Cluster)

Set up an Atlas cluster for the above Diabetes application. Connect your mongosh shell with the Atlas cluster and complete the following transactions (Note, each question should be completed in one individual transaction):

1. Randomly pick an employee, who has severe burnout risk, to record his/her stress level and the risk of burning out.
2. Randomly pick an employee, who has a good communication skill and can achieve goals over 70%, to record his/her collaboration score. According to his collaboration score and high project completion rate (above 70%), increase his/her salary by 2%.